Experiment No. 12 (VA 12)

TITLE :

Campus Event & Workshop Registration System – Interactive Real-Time Web Application using JavaScript and jQuery (DOM, Events, Web Storage and AJAX)

AIM :

To build an interactive real-time web application using JavaScript and jQuery with DOM manipulation, event handling, Web Storage and AJAX integration.

SOFTWARE / TOOLS REQUIRED :

- Visual Studio Code (VS Code)
- Google Chrome
- jQuery library

THEORY :

This experiment explores client-side computing by building a Campus Event & Workshop Registration System. The application lists campus workshops with live seat availability, lets students register for events, and keeps the roster, statistics and storage usage updated in real time without reloading the page.

1. **jQuery & DOM Manipulation:**
   jQuery selectors ($), .append(), .html(), .text(), .val() and .remove() are used to build event cards, populate the roster table and dropdowns, update seat counters and progress bars, and delete or add elements dynamically.
2. **Event Handling:**
   Click, change, input and submit events are attached with .on(), including delegated handlers for dynamically created buttons (Select Event, Remove, Delete). Search, filter and sort controls re-render the roster instantly.
3. **Form Validation:**
   The registration form checks that PRN, name, institutional email and event are filled, validates the email format, blocks duplicate PRNs for the same event and prevents booking when no seats remain.
4. **Web Storage (localStorage):**
   Events and registrations are saved as JSON under the key vac12_campus_data using setItem() / getItem(). Save, Reload, Clear and live storage-usage (KB) controls keep data persistent across page refreshes.
5. **AJAX Integration:**
   jQuery AJAX ($.ajax / $.getJSON) simulates server synchronisation to fetch the latest seat availability. A "Sync via AJAX" button and a selectable auto-refresh timer (setInterval) update the last-sync time and seat counts in real time.
6. **Real-Time Updates:**
   A live clock, next auto-check countdown, network status and seat tracking indicator are refreshed continuously. Deleting a registration releases the seat and increments capacity immediately.
7. **Data Operations:**
   Roster records can be searched by PRN, name or email, filtered by event, sorted (newest first, etc.) and exported as a CSV file; events can be added or removed dynamically.
8. **Responsive Layout:**
   CSS Grid/Flexbox with relative units and an academic white theme keeps cards, forms and tables readable on different screen sizes.

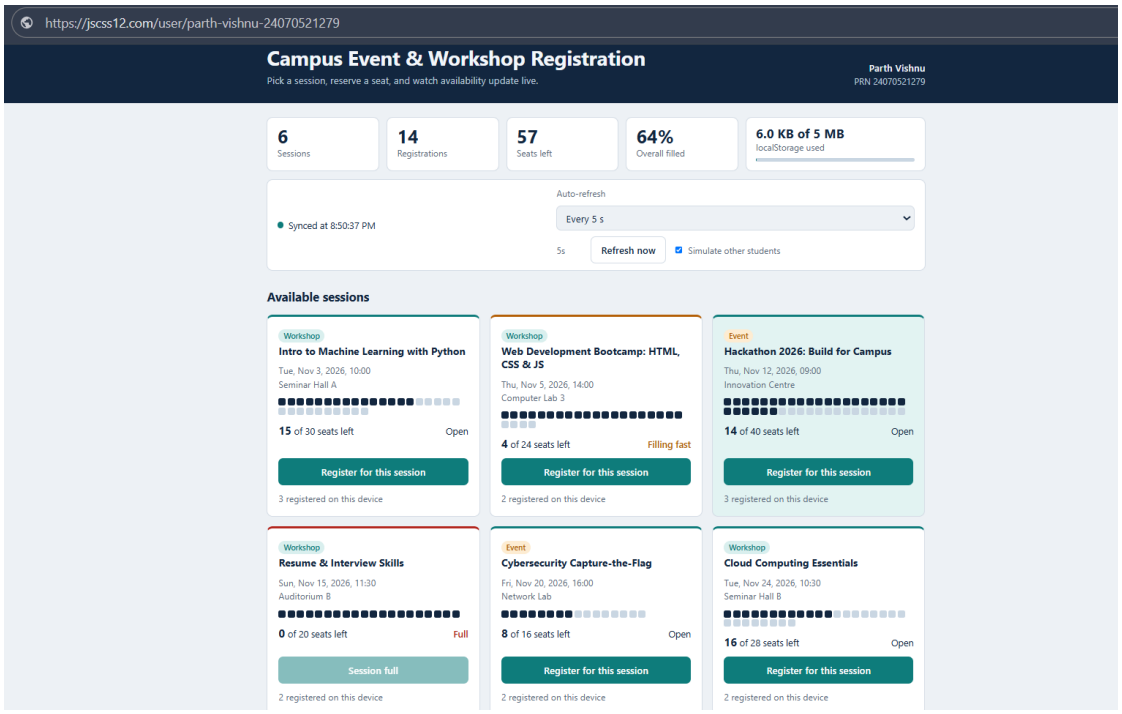
AIM OF THE CASE STUDY :

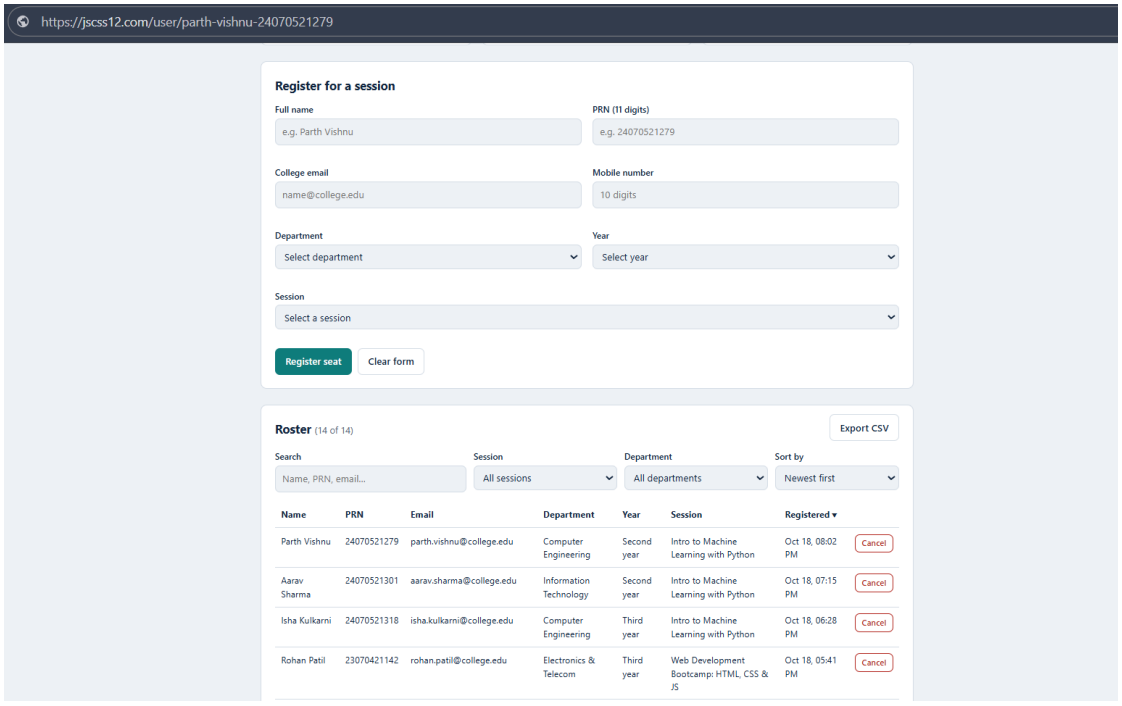
Develop a Campus Event & Workshop Registration System webpage that lists workshops with live seat availability, lets students register through a validated form, and keeps the roster, statistics and storage usage updated in real time. The page must use jQuery for DOM manipulation and event handling, save data in localStorage, synchronise seat availability through AJAX with a selectable auto-refresh timer, support roster search, filter, sort and CSV export, and remain responsive on different screen sizes.

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OUTPUT :





RESULT / CONCLUSION :

In conclusion, this experiment demonstrated the use of **JavaScript and jQuery** by developing a functional, real-time Campus Event & Workshop Registration System. jQuery DOM manipulation and event handling updated the interface dynamically, localStorage preserved events and registrations across reloads, and AJAX with a timed auto-refresh synchronised seat availability in real time. The registration form, roster search/filter/sort, CSV export and seat release on deletion worked correctly. Thus, the experiment demonstrated effective use of **DOM, events, Web Storage and AJAX** in an interactive web application.